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B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0603 OPERATIONS MANAGEMENT (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand different network techniques and forecasting methods

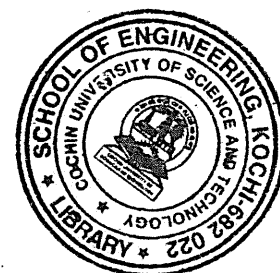
CO2: Learn production planning and control as well as inventory control techniques

CO3: Study the concepts of aggregate planning and scheduling

CO4: Get the basic knowledge of plant location and lay out and material handling

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome



PART A

(Answer *ALL* questions)

I.	(8 × 3 = 24)	Marks	BL	CO	PO
(a) What is critical Path in network analysis?		3	L1	1	1
(b) Differentiate between qualitative Vs quantitative forecasting.		3	L2	1	1
(c) Analyse how product life cycle is important in industry.		3	L4	2	1
(d) How Just in Time (JIT) manufacturing strategy helps in inventory control.		3	L3	2	2
(e) What is the need for aggregate planning?		3	L1	3	1
(f) Discuss critical ratio method of loading and scheduling of jobs to machines.		3	L1	3	2
(g) Identify factors that decides whether to go for maintenance of equipments or replacement.		3	L5	4	2
(h) Explain the concept of depreciation.		3	L1	4	1

PART B

(4 × 12 = 48)

II. (a)	Explain production process as a system and its various components briefly.	4	L2	1	1
(b)	Draw network of the following project and identify the critical path.	8	L3	1	2

Activity	A	B	C	D	E	F	G	H	I	J	K	L	M
Predecessors	-	A	B	A	D	E	-	G	J, H	-	A	C, K	I, L
Duration(Days)	6	4	7	2	4	10	2	10	6	13	9	3	5

Also, find out the project duration.

OR

(P.T.O)

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- III. (a) Generate the forecasted sale values of following time series data using a simple moving average (SMA) and with the weighted moving average (WMA) with averaging interval (m) = 3 for both cases. For WMA, the more recent observations are weighted more than older observations using the weights $w_1 = 0.6$, $w_2 = 0.3$ and $w_3 = 0.1$. 8 L6 1 2

t	Time Bucket (Month)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Y	Actual Sales in '000)	3	5	9	20	12	17	22	23	51	41	56	75	60	75	88

Compute the Mean Absolute Error (MAE) and Mean Square Error (MSE) of SMA and WMA. Comment on forecast obtained.

- (b) Discuss briefly: (i) slack (ii) multi-time estimate in PERT. 4 L1 1 1
- IV. (a) Distinguish briefly simplification, standardization, specialisation and inter-changeability. 6 L1 2 1
- (b) Purchasing department of a company estimates carrying cost as 15% and ordering cost at \$90 per order. The estimated annual requirement is 78,000 units at a price of \$4 per unit. 6 L3 2 2
- (i) What is the most economical number of units to order?
- (ii) Determine number of orders to be placed in a year.
- (iii) Find about how often will an order need to be placed.

OR

- V. (a) Discuss objectives and functions of PPC (Production Planning and Control). 6 L2 2 1
- (b) Explain EOQ model of inventory control. Derive the equation for EOQ. 6 L3 2 1
- VI. (a) Densepack, a disk drive producer has 300 workers employed at the moment. The ending inventory in December 500 units. The firm would like to have 600 units at the end of June. Inventory requirements and on-hand inventories are dealt with by adding/subtracting them to the expected demand. The expected demand based on forecast is as follows 8 L4 3 2

Demand and production data

Month ↓ Description →	Predicted demand	Net predicted demand	Net Cumulative demand	Number of Working days
January	1280	780	780	20
February	640	640	1420	24
March	900	900	2320	18
April	1200	1200	3520	26
May	2000	2000	5520	22
June	1400	2000	7520	15

Cost data

C_H : cost of hiring one worker = \$ 500; C_F : cost of firing one worker = \$ 1000;

C_I : cost of holding one unit of inventory one month = \$ 80; Salary to each worker: \$ 2500 per month. K ; Number of aggregate units produced by one worker in one day: It was observed that 76 workers produced 245 disk drives in 22; $K = 245/(76 \times 22) = 0.14653$

Strategy 1: Chase the demand by changing work force levels

Strategy 2: Keep the work force level constant throughout the planning horizon

Compare the total cost associated with above two Aggregate Planning Strategies. Draw graphs between months (X-axis) and Cumulative demand and production (Y-axis) for both plans and commend on the feasibility of the plans.

- (b) Explain Gantt Charts with sketches. 4 L2 3 1

OR

(Continued to 3)

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VII. (a)

JOB	A	B	C	D	E	F
Processing Time (t)	3	12	9	11	6	4
Due Date (D)	7	16	4	17	15	18

8 L3 3 2

The above independent jobs are sequenced into workstation with multiple job processing capacity from A to F. If jobs are sequenced under Short Processing Time (SPT), Due Dates (DD) and Critical ratio (CR) priority rules into the workstation, find the following for each rule.

1. Job Sequence
2. Total Tardiness
3. Total flow time
4. Average tardiness
5. Average flow time

(b) Explain job sequence using Johnson's rule?

4 L1 3 1

VIII. (a) Explain the following principles of Plant Layout briefly.

8 L2 4 1

- (i) Principle of Minimum Movement
- (ii) Principle of Space Utilization
- (iii) Principle of Flexibility
- (iv) Principle of Interdependence
- (v) Principle of Overall Integration
- (vi) Principle of Safety
- (vii) Principle of Smooth Flow
- (viii) Principle of Economy

(b) What are the factors to be considered for equipment replacement?

4 L2 4 1

OR

IX. (a) What is Systematic Plant Layout (SLP)? Explain the fundamental concepts for SLP

6 L1 4 1

(b) Explain briefly factors influencing plant Locations.

6 L2 4 1

Bloom's Taxonomy Levels

L1: 33.3%, L2: 29.2%, L3: 20.8%, L4: 8.3%, L5: 4.2%, L6: 4.2%
